

specguard report

specguard/examples/arp4754b_appendix_e.py

May 13, 2026

1 Encoded collections

PSSA_REQUIREMENTS

E12-1 Airplane electrical power 1 independent from airplane electrical power 2

$$\neg(\text{lost_elec_input_1} \wedge \text{lost_elec_input_2})$$

E12-2 NORMAL Mode braking function independent from ALTERNATE Mode braking function, for total loss failure condition

$$\neg(\text{failed_normal_braking} \wedge \text{failed_alternate_braking})$$

E12-3 The NMVs and their control are independent from the SOV and its control such that no single failure results in uncommanded braking

$$(\text{single_failure} \Rightarrow \neg \text{inadvertent_braking_takeoff})$$

E13-1 The probability of BSCU failure resulting in loss of a valid braking command output to the NMV shall not exceed 2.0E-04 per flight

$$p_{\text{bscu_loss_braking_cmd}} \leq 2 \times 10^{-4}$$

E13-2 The probability of BSCU failure resulting in unannunciated erroneous braking command to the NMV shall not exceed 2.0E-04 per flight

$$p_{\text{bscu_erroneous_cmd}} \leq 2 \times 10^{-4}$$

E13-3 The probability of BSCU failure resulting in the loss of command to open the SOV shall not exceed 2.0E-04 per flight

$$p_{\text{bscu_loss_sov_cmd}} \leq 2 \times 10^{-4}$$

E13-4 The probability of BSCU failure resulting in unintended closure of the S/ASV shall not exceed 2.0E-04 per flight

$$p_{\text{bscu_unintended_sasv}} \leq 2 \times 10^{-4}$$

E13-5 The SOV and NMV commands shall be provided by the BSCU upon loss of either airplane electrical power input

$$(((\text{lost_elec_input_1} \wedge \neg \text{lost_elec_input_2}) \Rightarrow (\text{bscu_provides_sov_cmd} \wedge \text{bscu_provides_nmv_cmd})) \wedge ((\text{lost_elec_input_2} \wedge \neg \text{lost_elec_input_1}) \Rightarrow (\text{bscu_provides_sov_cmd} \wedge \text{bscu_provides_nmv_cmd})))$$

E13-6 When "HYD 1 Enable" output is enabled, then "Alt/Emer Ctrl" output shall be disabled

$$(\text{hyd1_enable_on} \Rightarrow \neg \text{alt_emer_ctrl_on})$$

E13-7 When "HYD 1 Enable" output is disabled, then "Alt/Emer Ctrl" output shall be enabled

$$(\neg \text{hyd1_enable_on} \Rightarrow \text{alt_emer_ctrl_on})$$

E13-8 No single failure shall cause erroneous NMV command and inhibit the SOV function

$$\neg(\text{erroneous_nmv_cmd} \wedge \text{sov_function_inhibited})$$

E13-9 The wheel brake command function of the BSCU shall be developed to FDAL A

$$\text{bscu_cmd_fdal_a}$$

E14-1 The probability of 'Loss of Normal Braking System Hydraulic Equipment' will be less than 3.3E-05 per flight

$$p_{\text{loss_normal_hyd_equip}} < 3.3 \times 10^{-5}$$

E14-2 The probability of 'Loss of Alternate Braking System Hydraulic Equipment' will be less than 3.3E-05 per flight

$$p_{\text{loss_alt_hyd_equip}} < 3.3 \times 10^{-5}$$

E14-3 The probability of 7 or more wheel speed sensors erroneous or inoperative will be less than 1.0E-07 per flight

$$p_{\text{seven_or_more_sensors_per_flight}} < 1 \times 10^{-7}$$

E14-4 The probability of loss of an airplane electrical power bus will be less than 1.0E-04 per flight

$$p_{\text{loss_elec_bus_per_flight}} < 1 \times 10^{-4}$$

E14-5 The probability of loss of a Left brake pedal position input will be less than 1.0E-04 per flight

$$p_{\text{loss_pedal_per_flight}} < 1 \times 10^{-4}$$

E14-6 Airplane electrical power bus 1 is independent from airplane electrical power bus 2

$$\neg(\text{lost_elec_input_1} \wedge \text{lost_elec_input_2})$$

E14-7 HYD 1 hydraulic system is independent from HYD 2 hydraulic system

$$\neg(\text{failed_hyd_1} \wedge \text{failed_hyd_2})$$

ALT_PSSA_REQUIREMENTS

E28-WBS-ASMP-1 The probability of 'Loss of Normal Braking System Hydraulic Equipment' will be less than 3.3E-05 per flight

$$p_loss_normal_hyd_equip < 3.3 \times 10^{-5}$$

E28-WBS-ASMP-2 The probability of 'Loss of Alternate Braking System Hydraulic Equipment' will be less than 3.3E-05 per flight

$$p_loss_alt_hyd_equip < 3.3 \times 10^{-5}$$

E28-WBS-ASMP-3 The probability of 7 or more wheel speed sensors erroneous or inoperative will be less than 1.0E-07 per flight

$$p_seven_or_more_sensors_per_flight < 1 \times 10^{-7}$$

E28-WBS-ASMP-4 The failure rate for loss of an airplane electrical power bus will be less than 1.0E-04 per hour of flight

$$\lambda_{loss_elec_bus_per_hour} < 1 \times 10^{-4}$$

E28-WBS-ASMP-5 The failure rate for loss of a left brake pedal position input will be less than 1.0E-04 per hour of flight

$$\lambda_{loss_pedal_per_hour} < 1 \times 10^{-4}$$

E28-WBS-ASMP-6 Airplane electrical power bus 1 is independent from airplane electrical power bus 2

$$\neg(\text{lost_elec_input_1} \wedge \text{lost_elec_input_2})$$

E28-WBS-ASMP-7 HYD 1 hydraulic system is independent from HYD 2 hydraulic system

$$\neg(\text{failed_hyd_1} \wedge \text{failed_hyd_2})$$

SPEC_REQUIREMENTS

S18-WBS-R-0020 The Wheel Brake System shall have a means to decelerate the wheels on the ground

$$wbs_has_decelerate_means$$

S18-WBS-R-0021 The Wheel Brake System shall be capable of decelerating the S18 airplane to a complete stop/to taxi speed in 2000 feet when wheel brakes, high lift speed brakes and reverse thrust are available, including when at maximum landing weight

$$stopping_distance_ft \leq 2000$$

S18-WBS-R-0041 The Wheel Brake System shall provide directional control on ground by a differential braking function

$$wbs_differential_braking$$

S18-WBS-R-0042 The Wheel Brake System shall provide a parking brake to prevent airplane motion on the ground

wbs_parking_brake

S18-WBS-R-0043 The Wheel Brake System shall provide autobraking capability during landing and RTO

wbs_autobraking

S18-WBS-R-0044 The Wheel Brake System shall provide anti-skid braking

wbs_antiskid

S18-WBS-R-0045 The Wheel Brake System shall provide hydraulic brake control

wbs_hyd_brake_control

S18-WBS-R-0046 The Wheel Brake System shall override the autobrake when the commanded by the flight crew

wbs_overrides_autobrake_on_crew

S18-WBS-R-0047 The Wheel Brake System shall meet safety requirements while operating in an average atmospheric radiation environment per the IEC 62396 standard with an altitude of 40000 feet and a latitude of 45 degrees

wbs_radiation_qualified

S18-WBS-R-0049 The Wheel Brake System shall be controlled and monitored by a computer system called Brake System Control Unit

wbs_controlled_by_bscu

S18-WBS-R-0050 Each wheel brake shall have separate hydraulic power supply lines

each_wheel_separate_hyd_lines

S18-WBS-R-0052 Each hydraulic Wheel Brake System circuit shall have meter valves, anti-skid valves and hydraulic fuses

each_circuit_valves_fuses

S18-WBS-R-0055 Anti-skid system shall be capable to prevent skidding of the tires by reducing the pressure applied to the brakes

antiskid_prevents_skidding

S18-WBS-R-0062 The Wheel Brake System shall include an emergency accumulator to supply hydraulic power to the wheel brakes

has_emergency_accumulator

S18-WBS-R-0065 The emergency accumulator shall provide a minimum of 1800 psi

$$\text{accumulator_pressure_psi} \geq 1800$$

S18-WBS-R-0066 The Alternate/Emergency Brake System hydraulic equipment and piping shall be installed aft of the engine 1 UERF trajectory envelope

$$\text{alt_emer_aft_of_uerf}$$

S18-WBS-R-0100 The Wheel Brake System decelerate the wheels on the ground function shall be developed as FDAL A

$$\text{wbs_decelerate_fdal_a}$$

S18-WBS-R-0120 No single failure or event shall cause the complete loss of hydraulic power for both the WBS Normal and Alternate/Emergency Brake Systems

$$\neg(\text{failed_normal_braking} \wedge \text{failed_alternate_braking})$$

S18-WBS-R-0130 No single failure or event shall cause the complete loss of electrical power for both the WBS Normal and Alternate/Emergency Brake Systems

$$\neg(\text{lost_elec_input_1} \wedge \text{lost_elec_input_2})$$

S18-WBS-R-0150 Complete loss of decelerate the wheels on the ground function shall be less probable than 1.0E-07 for a landing

$$\text{p_total_loss_decelerate} < 1 \times 10^{-7}$$

S18-WBS-R-0200 Two redundant control lanes shall be provided between the EBU and each of the two Alternate/Emergency Meter Valves

$$\text{ebu_two_redundant_lanes}$$

S18-WBS-R-0326 No single failure shall result in inadvertent wheel braking of all wheels during takeoff roll

$$(\text{single_failure} \Rightarrow \neg \text{inadvertent_braking_takeoff})$$

S18-WBS-R-0508 The Wheel Brake System shall have at least two independent hydraulic pressure sources

$$\text{wbs_two_independent_hyd_sources}$$

S18-WBS-R-0509 The Wheel Brake System shall have dual BSCU command functions

$$\text{wbs_dual_bscu_cmd}$$

S18-WBS-R-0510 The rudder and nose wheel steering functions shall not be implemented by the Wheel Brake System

$$\text{wbs_no_rudder_or_nws}$$

S18-WBS-R-0631 The Wheel Brake System shall indicate individual brake temperature

wbs_indicates_brake_temp

S18-WBS-R-0632 The Wheel Brake System shall control individual brake pressure

wbs_individual_brake_pressure

S18-WBS-ICD-1002 The brake actuators shall respond to a BSCU command within 5 ms of receiving it

actuator_response_time_ms $\leq$ 5

S18-WBS-R-1613 Hydraulic pressure shall be controlled for weight, autobrake mode, ground speed, wheel rotation, brake temperature and deceleration rate in accordance with graphs given in S18 brake force analysis AAS18-XXX

hyd_pressure_per_analysis

S18-WBS-R-2971 Wheel Brake System shall have one NORMAL operating mode

wbs_one_normal_mode

S18-WBS-R-2972 Wheel Brake System shall have one ALTERNATE operating mode

wbs_one_alternate_mode

S18-WBS-R-2973 Operation of the Alternate/Emergency Brake System shall be precluded when the Normal Brake System is in use

(normal_mode_active $\Rightarrow$ $\neg$ alternate_mode_active)

S18-WBS-R-2974 An emergency accumulator shall provide hydraulic power for EMERGENCY Wheel Brake System operating mode

accumulator_powers_emergency

S18-WBS-R-2975 The emergency accumulator shall be attached to the HYD 2 hydraulic line between the Selector Valve and the Alternate/Emergency Meter Valve

accumulator_attached_to_hyd2

S18-WBS-R-2976 Wheel Brake System shall charge the emergency accumulator prior to normal operation

wbs_charges_accumulator_pre_normal

S18-WBS-R-2986 The wheel brake command function of the BSCU shall be developed as FDAL A

bscu_cmd_fdal_a

S18-WBS-R-2997 The BSCU shall have two independent command channels

bscu_two_independent_channels

S18-WBS-R-2999 The BSCU shall have two independent electrical power sources

bscu_two_independent_power

S18-WBS-R-3011 Brake pedals shall be independent from the rudder control

brake_pedals_indep_rudder

S18-WBS-R-3213 Normal Brake System shall be powered from the airplane's HYD 1 hydraulic system

(failed_hyd_1 $\Rightarrow$ failed_normal_braking)

S18-WBS-R-3224 Alternate/Emergency Brake System shall be powered from the airplane's HYD 2 hydraulic system

(failed_hyd_2 $\Rightarrow$ failed_alternate_braking)

S18-WBS-R-3245 ALTERNATE Mode shall be automatically selected when the NORMAL Mode fails

(normal_mode_failed $\Rightarrow$ alternate_mode_active)

S18-WBS-R-6104 The probability of BSCU failure resulting in loss of a valid braking command output to the NMV shall not exceed 2.0E-04 per flight

$$p_bscu_loss_braking_cmd \leq 2 \times 10^{-4}$$

S18-WBS-R-6105 The probability of BSCU failure resulting in unannunciated erroneous braking command to the NMV shall not exceed 2.0E-04 per flight

$$p_bscu_erroneous_cmd \leq 2 \times 10^{-4}$$

S18-WBS-R-6106 The probability of BSCU failure resulting in the loss of command to open the SOV shall not exceed 2.0E-04 per flight

$$p_bscu_loss_sov_cmd \leq 2 \times 10^{-4}$$

S18-WBS-R-6107 The probability of BSCU failure resulting in unintended closure of the S/ASV shall not exceed 2.0E-04 per flight

$$p_bscu_unintended_sasv \leq 2 \times 10^{-4}$$

S18-WBS-R-6108 The SOV and NMV commands shall be provided by the BSCU upon loss of either airplane electrical power input

$(((\text{lost_elec_input_1} \wedge \neg \text{lost_elec_input_2}) \Rightarrow (\text{bscu_provides_sov_cmd} \wedge \text{bscu_provides_nmv_cmd})) \wedge ((\text{lost_elec_input_2} \wedge \neg \text{lost_elec_input_1}) \Rightarrow (\text{bscu_provides_sov_cmd} \wedge \text{bscu_provides_nmv_cmd})))$

S18-WBS-R-6109 When "HYD 1 Enable" output is enabled, then "Alt/Emer Ctrl" output shall be disabled

$$(\text{hyd1_enable_on} \Rightarrow \neg \text{alt_emer_ctrl_on})$$

S18-WBS-R-6110 No single failure shall cause erroneous NMV command and inhibit the SOV function

$$\neg(\text{erroneous_nmv_cmd} \wedge \text{sov_function_inhibited})$$

S18-WBS-R-6111 The probability of 'Loss of Normal Brake System Hydraulic Components' shall be less than $3.3\text{E-}05$ per flight

$$p_loss_normal_hyd_equip < 3.3 \times 10^{-5}$$

S18-WBS-R-6112 The probability of 'Loss of Alternate/Emergency Brake System Hydraulic Components' shall be less than $3.3\text{E-}05$ per flight

$$p_loss_alt_hyd_equip < 3.3 \times 10^{-5}$$

BSCU_PSSA_REQUIREMENTS

BSCU-001 Channel #1 shall have physical independence from Channel #2

$$\text{channel1_phys_indep_channel2}$$

BSCU-002 Within each channel, the command function shall have physical independence from the monitor function

$$\text{cmd_phys_indep_monitor}$$

BSCU-003 The COMX software and MONX software shall be developed using different software requirement specifications

$$\text{comx_monx_diff_specs}$$

BSCU-004 Within each channel, a power supply monitor shall shutoff the power supply when any voltage is detected to be out of specification

$$(\text{voltage_out_of_spec} \Rightarrow \text{power_supply_shutoff})$$

BSCU-005 Within each channel, the operational state of the power supply shall have no effect on the operational state of the power supply monitor

$$\text{power_supply_state_indep_monitor}$$

BSCU-006 Upon Channel #1 indicating an 'invalid' state, Channel #2 shall command the 'Normal Control' output

$$(\text{channel1_invalid} \Rightarrow \text{channel2_cmd_normal_ctrl})$$

BSCU-007 Upon both Channel #1 and Channel #2 indicating 'invalid' states, the 'HYD 1 Enable' output shall be disabled

$$((\text{channel1_invalid} \wedge \text{channel2_invalid}) \Rightarrow \neg \text{hyd1_enable_on})$$

BSCU-008 Upon either Channel #1 or Channel #2 indicating a validity error, a WBS annunciation shall be output from the BSCU

$$((\text{channel1_invalid} \vee \text{channel2_invalid}) \Rightarrow \text{wbs_annunciation_out})$$

BSCU-009 At power-up the BSCU shall perform a self-test of Channel #2 including the ability to switch control from Channel #1 to Channel #2

$$\text{power_up_self_tests_ch2}$$

BSCU-010 At power-up each channel of the BSCU shall implement a test of its power supply monitor

$$\text{power_up_tests_psm}$$

BSCU-011 If the power-up test fails, a WBS annunciation shall be output from the BSCU

$$(\text{power_up_test_failed} \Rightarrow \text{wbs_annunciation_out})$$

BSCU_ASSUMPTIONS_TO_WBS

E23-1 When 'HYD 1 Enable' is disabled, the BSCU brake control outputs are not used

$$(\neg \text{hyd1_enable_on} \Rightarrow \neg \text{bscu_brake_outputs_in_use})$$

E23-2 A BSCU maintenance action will be initiated, if the WBS Status indicates a failure

$$(\text{wbs_status_failure} \Rightarrow \text{bscu_maintenance_initiated})$$

BSCU_SPEC_REQUIREMENTS

S18-BSCU-R-0001 The wheel brake command function of the BSCU shall be developed as FDAL A

$$\text{bscu_cmd_fdal_a}$$

S18-BSCU-R-0002 The probability of BSCU failure resulting in loss of a valid braking command output to the NMV shall not exceed 2.0E-04 per flight

$$p_{\text{bscu_loss_braking_cmd}} \leq 2 \times 10^{-4}$$

S18-BSCU-R-0003 The probability of BSCU failure resulting in unannunciated erroneous braking command to the NMV shall not exceed 2.0E-04 per flight

$$p_{\text{bscu_erroneous_cmd}} \leq 2 \times 10^{-4}$$

S18-BSCU-R-0004 The probability of BSCU failure resulting in the loss of command to open the SOV shall not exceed 2.0E-04 per flight

$$p_{\text{bscu_loss_sov_cmd}} \leq 2 \times 10^{-4}$$

S18-BSCU-R-0005 The probability of BSCU failure resulting in unintended closure of the S/ASV shall not exceed 2.0E-04 per flight

$$p_bscu_unintended_sasv \leq 2 \times 10^{-4}$$

S18-BSCU-R-0006 The SOV and NMV commands shall be provided by the BSCU upon loss of either airplane electrical power input

$$(((lost_elec_input_1 \wedge \neg lost_elec_input_2) \Rightarrow (bscu_provides_sov_cmd \wedge bscu_provides_nmv_cmd)) \wedge ((lost_elec_input_2 \wedge \neg lost_elec_input_1) \Rightarrow (bscu_provides_sov_cmd \wedge bscu_provides_nmv_cmd)))$$

S18-BSCU-R-0007 When "HYD 1 Enable" output is enabled, then "Alt/Emer Ctrl" output shall be disabled

$$(hyd1_enable_on \Rightarrow \neg alt_emer_ctrl_on)$$

S18-BSCU-R-0008 No single failure shall cause erroneous NMV command and inhibit the SOV function

$$\neg (erroneous_nmv_cmd \wedge sov_function_inhibited)$$

S18-BSCU-R-0009 When 'HYD 1 Enable' is disabled, the BSCU brake control outputs shall be disabled

$$(\neg hyd1_enable_on \Rightarrow \neg bscu_brake_outputs_in_use)$$

S18-BSCU-R-0010 BSCU shall calculate the required hydraulic pressure as a function of weight, autobrake mode, ground speed, wheel rotation, brake temperature and deceleration rate in accordance with graphs given in S18 brake force analysis AAS18-XXX

$$hyd_pressure_per_analysis$$

S18-BSCU-R-0011 BSCU shall control the meter valves

$$bscu_controls_meter_valves$$

S18-BSCU-R-0012 BSCU shall control the anti-skid valves

$$bscu_controls_antiskid_valves$$

S18-BSCU-R-0013 BSCU shall control the shut off valves

$$bscu_controls_shutoff_valves$$

S18-BSCU-R-0020 BSCU shall control the valves to charge the emergency accumulator prior to normal operation

$$bscu_charges_accumulator_pre_normal$$

S18-BSCU-R-0027 Each BSCU shall have two command channels

$$bscu_each_two_command_channels$$

S18-BSCU-R-0051 Each BSCU channel shall have a monitoring unit

bscu_each_channel_has_monitoring

S18-BSCU-R-0075 Each BSCU channel shall have a control unit

bscu_each_channel_has_control

S18-BSCU-R-0099 Each BSCU channel shall have independent power

bscu_each_channel_indep_power

S18-BSCU-R-0123 Each BSCU channel shall have independent pedal input to both control and monitoring units

bscu_each_channel_indep_pedal

S18-BSCU-R-0124 The command channels of each BSCU shall be independent of each other

bscu_command_channels_indep

S18-BSCU-R-0126 The BSCU hardware shall be developed to IDAL A

bscu_hw_idal_a

S18-BSCU-R-0127 The BSCU software shall be developed to IDAL B

bscu_sw_idal_b

S18-BSCU-R-0128 BSCU shall meet safety requirements while operating in an average atmospheric radiation environment per the IEC 62396 standard with an altitude of 40000 feet and a latitude of 45 degrees

bscu_radiation_qualified

S18-BSCU-R-0201 Channel #1 shall have physical independence from Channel #2

channel1_phys_indep_channel2

S18-BSCU-R-0202 Within each channel, the command function shall have physical independence from the monitor function

cmd_phys_indep_monitor

S18-BSCU-R-0203 The COMX software and MONX software shall be developed using different software requirement specifications

comx_monx_diff_specs

S18-BSCU-R-0204 Within each channel, a power supply monitor shall shutoff the power supply when any voltage is detected to be out of specification

(voltage_out_of_spec $\Rightarrow$ power_supply_shutoff)

S18-BSCU-R-0205 Within each channel, the operational state of the power supply shall have no effect on the operational state of the power supply monitor

power_supply_state_indep_monitor

S18-BSCU-R-0206 Upon Channel #1 indicating an 'invalid' state, Channel #2 shall command the 'Normal Control' output

$(\text{channel1_invalid} \Rightarrow \text{channel2_cmd_normal_ctrl})$

S18-BSCU-R-0207 Upon both Channel #1 and Channel #2 indicating 'invalid' states, the 'HYD 1 Enable' output shall be disabled

$((\text{channel1_invalid} \wedge \text{channel2_invalid}) \Rightarrow \neg \text{hyd1_enable_on})$

S18-BSCU-R-0208 Upon either Channel #1 or Channel #2 indicating a validity error, a WBS annunciation shall be output from the BSCU

$((\text{channel1_invalid} \vee \text{channel2_invalid}) \Rightarrow \text{wbs_annunciation_out})$

S18-BSCU-R-0209 At power-up, the BSCU shall perform a self-test of Channel #2 including the ability to switch control from Channel #1 to Channel #2

power_up_self_tests_ch2

S18-BSCU-R-0210 At power-up, each channel of the BSCU shall implement a test of its power supply monitor

power_up_tests_psm

S18-BSCU-R-0211 If the power-up test fails, a WBS annunciation shall be output from the BSCU

$(\text{power_up_test_failed} \Rightarrow \text{wbs_annunciation_out})$

AIRPLANE_REQUIREMENTS

S18-ACFT-R-1000 The S18 airplane shall have a means to decelerate on ground

airplane_decelerate_on_ground

S18-ACFT-R-1100 The S18 airplane shall have a means to decelerate the wheels on ground

airplane_decelerate_wheels_on_ground

S18-ACFT-R-1110 The S18 airplane shall have pilot-controlled wheel braking capability

airplane_pilot_controlled_braking

S18-ACFT-R-1120 The S18 airplane shall have an autobraking capability during landing and RTO

airplane_autobraking

S18-ACFT-R-1130 The S18 airplane shall have anti-skid braking

airplane_antiskid

S18-ACFT-R-1140 The S18 airplane shall provide interface for Wheel Brake System status and annunciations

airplane_wbs_status_interface

S18-ACFT-R-0181 The S18 airplane shall decelerate the wheels on gear retraction

airplane_decelerate_wheels_on_gear_retract

S18-ACFT-R-0182 The S18 airplane shall be capable of decelerating the wheels differentially

airplane_decelerate_differentially

S18-ACFT-R-0183 The S18 airplane shall have a parking brake to prevent airplane motion when parked

airplane_parking_brake

S18-ACFT-R-0184 The S18 airplane shall have hydraulically driven braking

airplane_hydraulic_braking

S18-ACFT-R-0185 The S18 airplane shall have an autobrake override that can be initiated by the flight crew

airplane_autobrake_override

S18-ACFT-R-0186 The S18 airplane shall meet safety requirements while operating in an average atmospheric radiation environment per the IEC 62396 standard with an altitude of 40000 feet and a latitude of 45 degrees

airplane_radiation_qualified

S18-ACFT-R-0933 The Wheel Brake System decelerate the wheels on the ground function shall be developed as FDAL A

airplane_decelerate_function_fdal_a

S18-ACFT-R-1322 No single failure or event shall result in the complete loss of wheel brake and the loss of one thrust reverser

$\neg(\text{complete_loss_wheel_brake} \wedge \text{loss_one_thrust_reverser})$

S18-ACFT-R-1385 Complete loss of wheel braking shall be less than 1.0E-07 for a landing

$p_{\text{complete_loss_wheel_braking_per_landing}} < 1 \times 10^{-7}$

S18-ACFT-R-1550 Loss of power from both hydraulic subsystems powered by the engines shall not lead to complete loss of wheel braking

$$(\text{both_engine_hyd_failed} \Rightarrow \neg \text{complete_loss_wheel_brake})$$

S18-ACFT-R-1551 Wheel Brake System shall include an emergency accumulator to supply hydraulic power to the wheel brakes

$$\text{airplane_has_emergency_accumulator}$$

S18-ACFT-R-1552 The Alternate/Emergency Brake System hydraulic equipment and piping shall be installed aft of the engine 1 UERF trajectory envelope

$$\text{alt_emer_aft_of_uerf}$$

S18-ACFT-R-1600 Two redundant control lanes shall be provided between the Electric Brake Unit (EBU) and each of the two Alternate/Emergency Meter Valves

$$\text{ebu_two_redundant_lanes}$$

PASA_SAFETY_REQUIREMENTS

PASA-SR-01 Decelerate wheels function shall be developed FDAL A

$$\text{airplane_decelerate_function_fdal_a}$$

PASA-SR-05 [FF1.1] Complete loss of wheel brake shall be less than $1.0\text{E-}07$ for a landing.

$$\text{p_complete_loss_wheel_braking_per_landing} < 1 \times 10^{-7}$$

PASA-SR-10 No single failure or event shall result in the complete loss of wheel brake and the loss of one thrust reverser.

$$\neg(\text{complete_loss_wheel_brake} \wedge \text{loss_one_thrust_reverser})$$

PASA-SR-12 Loss of power from both hydraulic subsystems powered by the engines shall not lead to complete loss of wheel braking.

$$(\text{both_engine_hyd_failed} \Rightarrow \neg \text{complete_loss_wheel_brake})$$

PASA-SR-13 The Alternate/Emergency Brake System hydraulic equipment and piping shall be installed aft of the engine 1 UERF trajectory envelope.

$$\text{alt_emer_aft_of_uerf}$$

PASA-SR-14 Two redundant control lanes shall be provided between the Electric Brake Unit and each of the two Alternate/Emergency Meter Valves.

$$\text{ebu_two_redundant_lanes}$$

AIRPLANE_ASSUMPTIONS

ASMP 3.2.2-1 Overrunning the runway length above "XYZ" knots is considered a high speed overrun.

$$(\text{groundspeed_overrun_kt} > \text{xyz_overrun_threshold_kt} \Rightarrow \text{high_speed_overrun})$$

ASMP 3.2.2-2 The directional aspect of asymmetric failures of deceleration systems are functionally addressed by failure conditions of the "Control direction on ground" airplane function (see function 2.3 AFHA).

$$\text{afha_dir_addressed_by_ctrl_dir}$$

ASMP 3.2.2-3 The flight crew will divert to a suitable airfield if aware of a condition that renders the airplane incapable of landing at the originally intended destination.

$$\text{crew_diverts_when_aware_unlandable}$$

ASMP 3.2.2-4 Crew awareness is not a factor for the identified malfunction, as these are immediately evident due to aircraft behavior and do not have their effects intensified or mitigated by crew awareness features.

$$\text{malfunction_immediately_evident}$$

ASMP 3.2.2-5 The flight crew will not initiate a Rejected Takeoff (RTO) in response to an annunciated failure of deceleration features during takeoff due to alert suppression.

$$(\text{annunciated_failure_decel_takeoff} \Rightarrow \neg \text{rto_initiated})$$

ASMP 3.2.2-6 Taxi is performed at groundspeeds below 30 knots.

$$\text{groundspeed_taxi_kt} < 30$$

ASMP 3.2.2-7 Landings with failure condition in combination with environmental factors have been assessed.

$$\text{landings_with_env_factors_assessed}$$

ASMP 3.2.2-8 Failures of deceleration capability will be detected and annunciated by on-board systems.

$$(\text{failure_decel_capability} \Rightarrow \text{detected_and_annunciated_onboard})$$

PASA-ASMP-01 It is assumed that the high speed overrun is above 30 knots and low speed overrun is below (or equal) 30 knots. This assumption has been derived from the AFHA assumption ASMP 3.2.2-1 and ASMP 3.2.2-6 for the establishment of the criteria and terms of "high-speed overrun" and "low speed overrun."

$$\begin{aligned} &(\text{high_speed_overrun} = \text{groundspeed_overrun_kt} > 30 \\ &\quad \wedge \text{low_speed_overrun} = \text{groundspeed_overrun_kt} \leq 30) \end{aligned}$$

PASA-ASMP-02 The degraded state of systems considered in the CoFFE analysis has been defined such as half functional capability.

degraded_state_half_capability

PASA-ASMP-03 Wheel Brake System uses ground detection information together with the wheel speed information. These functions are developed independently so that even if Erroneous Ground Detection Information (AGS.MF) is provided on ground, i.e., with false in-flight status, wheel braking function is available if there is correct wheel speed information.

((erroneous_ground_detection
 $\wedge$ correct_wheel_speed_info) $\Rightarrow$ wheel_braking_function_available)

PRA-UERF-ASSUMPTION-01 The BSCU is installed in the avionics compartment located in the nose fuselage section of the airplane, forward of the UERF area.

bscu_in_avionics_compartment_fwd

PRA-UERF-ASSUMPTION-02 The electrical lines transmitting the brake control signals from the Electric Brake Unit to the BSCU are routed "to the shortest" from one end to the other in the nose fuselage section of the airplane, thus are kept forward of the UERF area.

ebu_to_bscu_lines_routed_short

PRA-UERF-ASSUMPTION-03 The hydraulic reservoirs and the high pressure manifolds of the hydraulic subsystems 1 and 2 are located in the fuselage, aft of the UERF area.

hyd_reservoirs_aft_of_uerf

PRA-UERF-ASSUMPTION-04 The hydraulic power distribution lines from the engine driven pumps to the respective hydraulic reservoirs and high pressure manifolds exit the engine pylons aft of the wing rear spars, then run along, aft of, the wing rear spars to the fuselage, and are kept aft of the UERF area inside the fuselage.

hyd_distribution_aft_of_uerf

PRA-UERF-ASSUMPTION-05 The hydraulic distribution lines routed to the forward part of the fuselage to supply hydraulic equipment located forward of the UERF area are fitted with appropriate isolation means located aft of the UERF area.

hyd_forward_lines_isolated

PRA-UERF-ASSUMPTION-06 The power supply lines from the electrical power center(s) to the EBU are routed "to the shortest" from one end to the other in the nose fuselage section of the airplane.

power_supply_lines_routed_short

PRA-UERF-ASSUMPTION-07 In case of cabin depressurization the airplane altitude will be limited by procedure to 10000 feet.

cabin_depress_alt_limited_10000ft

WBS_SFHA_ASSUMPTIONS

SASP 1.1-1 Loss of 80% or more of deceleration capability is considered a total loss of the deceleration means.

$$(\text{deceleration_capability_loss_pct} \geq 80 \Rightarrow \text{total_loss_deceleration})$$

SASP 1.1-2 Failure Conditions are assessed on wet runway conditions.

$$\text{failure_conditions_assessed_wet}$$

SASP 1.1-3 RTO is considered an operational condition caused either by an external event or by a system failure (annunciated or perceived by the flight crew) during takeoff run.

$$\text{rto_caused_external_or_failure}$$

SASP 1.1-4 The flight crew will not initiate an RTO due to an annunciated failure of the deceleration function after V1.

$$(\text{annunciated_failure_decel_after_v1} \Rightarrow \neg \text{rto_initiated})$$

SASP 1.1-5 Taxi is performed at groundspeeds below 30 knots.

$$\text{groundspeed_taxi_kt} < 30$$

SASP 1.1-6 Overrunning the runway length at or above "XYZ" knots is considered a high speed overrun.

$$(\text{groundspeed_overrun_kt} \geq \text{xyz_overrun_threshold_kt} \Rightarrow \text{high_speed_overrun})$$

SASP 1.1-7 Overrunning the runway length below "XYZ" knots is considered a low speed overrun.

$$(\text{groundspeed_overrun_kt} < \text{xyz_overrun_threshold_kt} \Rightarrow \text{low_speed_overrun})$$

SASP 1.1-8 Failures of deceleration capability will be detected and annunciated by on-board systems.

$$(\text{failure_decel_capability} \Rightarrow \text{detected_and_annunciated_onboard})$$

AIRPLANE_CATASTROPHIC_OBJECTIVES

E.3.11-3 The airplane does not satisfy the safety objective for loss of deceleration capability if the probability of loss of both HYD 1 and HYD 2 was not extremely improbable ($>1.0\text{E-}09$).

$$\text{p_loss_both_hyd_per_flight} \leq \text{catastrophic_loss_decel_threshold}$$

2 Axioms

$$\text{flight_duration_hours} = 5$$

$$\text{p_loss_elec_bus_per_flight} = \text{lambda_loss_elec_bus_per_hour} \cdot \text{flight_duration_hours}$$

$$\text{p_loss_pedal_per_flight} = \text{lambda_loss_pedal_per_hour} \cdot \text{flight_duration_hours}$$

$$\text{p_loss_both_hyd_per_flight} = \text{p_loss_normal_hyd_equip} \cdot \text{p_loss_alt_hyd_equip}$$

$$\text{catastrophic_loss_decel_threshold} = 1 \times 10^{-9}$$

3 Internal consistency

- PSSA_REQUIREMENTS: SAT
- ALT_PSSA_REQUIREMENTS: SAT
- SPEC_REQUIREMENTS: SAT
- BSCU_PSSA_REQUIREMENTS: SAT
- BSCU_ASSUMPTIONS_TO_WBS: SAT
- BSCU_SPEC_REQUIREMENTS: SAT
- AIRPLANE_REQUIREMENTS: SAT
- PASA_SAFETY_REQUIREMENTS: SAT
- AIRPLANE_ASSUMPTIONS: SAT
- WBS_SFHA_ASSUMPTIONS: SAT
- AIRPLANE_CATASTROPHIC_OBJECTIVES: SAT

4 Directional entailment

AIRPLANE_REQUIREMENTS entails each PASA_SAFETY_REQUIREMENTS

- PASA-SR-01: PASS
- PASA-SR-05: PASS
- PASA-SR-10: PASS
- PASA-SR-12: PASS
- PASA-SR-13: PASS
- PASA-SR-14: PASS

Summary: 6 passed, 0 failed.

SPEC_REQUIREMENTS entails each PSSA_REQUIREMENTS

- E12-1: PASS
- E12-2: PASS
- E12-3: PASS
- E13-1: PASS
- E13-2: PASS
- E13-3: PASS
- E13-4: PASS
- E13-5: PASS
- E13-6: PASS
- E13-7: FAIL
text: When "HYD 1 Enable" output is disabled, then "Alt/Emer Ctrl" output shall be enabled
witness: alt_emer_ctrl_on=False, hyd1_enable_on=False
- E13-8: PASS
- E13-9: PASS
- E14-1: PASS
- E14-2: PASS
- E14-3: FAIL
text: The probability of 7 or more wheel speed sensors erroneous or inoperative will be less than 1.0E-07 per flight
witness: p_seven_or_more_sensors_per_flight=1.0000e-07
- E14-4: FAIL
text: The probability of loss of an airplane electrical power bus will be less than 1.0E-04 per flight
witness: p_loss_elec_bus_per_flight=1.0000e-04
- E14-5: FAIL
text: The probability of loss of a Left brake pedal position input will be less than 1.0E-04 per flight
witness: p_loss_pedal_per_flight=1.0000e-04
- E14-6: PASS
- E14-7: PASS

Summary: 15 passed, 4 failed.

BSCU_SPEC_REQUIREMENTS entails each BSCU_PSSA_REQUIREMENTS

- BSCU-001: PASS
- BSCU-002: PASS

- BSCU-003: PASS
- BSCU-004: PASS
- BSCU-005: PASS
- BSCU-006: PASS
- BSCU-007: PASS
- BSCU-008: PASS
- BSCU-009: PASS
- BSCU-010: PASS
- BSCU-011: PASS

Summary: 11 passed, 0 failed.

PSSA_REQUIREMENTS entails each AIRPLANE_CATASTROPHIC_OBJECTIVES

- **E.3.11-3: FAIL**
text: The airplane does not satisfy the safety objective for loss of deceleration capability if the probability of loss of both HYD 1 and HYD 2 was not extremely improbable ($>1.0E-09$).
witness: catastrophic_loss_decel_threshold=1.0000e-09, p_loss_both_hyd_per_flight=1.0041e-0

Summary: 0 passed, 1 failed.

5 Restatement equivalence

- **E14-1 $\Leftrightarrow$ E28-WBS-ASMP-1: EQUIVALENT**
- **E14-2 $\Leftrightarrow$ E28-WBS-ASMP-2: EQUIVALENT**
- **E14-3 $\Leftrightarrow$ E28-WBS-ASMP-3: EQUIVALENT**
- **E14-4 $\Leftrightarrow$ E28-WBS-ASMP-4: NOT EQUIVALENT**
E14-4: The probability of loss of an airplane electrical power bus will be less than $1.0E-04$ per flight
E28-WBS-ASMP-4: The failure rate for loss of an airplane electrical power bus will be less than $1.0E-04$ per hour of flight
E14-4 entails E28-WBS-ASMP-4: PASS
E28-WBS-ASMP-4 entails E14-4: FAIL; witness: p_loss_elec_bus_per_flight=4.8828e-04
- **E14-5 $\Leftrightarrow$ E28-WBS-ASMP-5: NOT EQUIVALENT**
E14-5: The probability of loss of a Left brake pedal position input will be less than $1.0E-04$ per flight
E28-WBS-ASMP-5: The failure rate for loss of a left brake pedal position input will be less than $1.0E-04$ per hour of flight
E14-5 entails E28-WBS-ASMP-5: PASS
E28-WBS-ASMP-5 entails E14-5: FAIL; witness: p_loss_pedal_per_flight=4.8828e-04
- **E14-6 $\Leftrightarrow$ E28-WBS-ASMP-6: EQUIVALENT**
- **E14-7 $\Leftrightarrow$ E28-WBS-ASMP-7: EQUIVALENT**

- ASMP 3.2.2-6 $\Leftrightarrow$ SASP 1.1-5: EQUIVALENT
- ASMP 3.2.2-8 $\Leftrightarrow$ SASP 1.1-8: EQUIVALENT
- ASMP 3.2.2-1 $\Leftrightarrow$ SASP 1.1-6: NOT EQUIVALENT
ASMP 3.2.2-1: Overrunning the runway length above "XYZ" knots is considered a high speed overrun.
SASP 1.1-6: Overrunning the runway length at or above "XYZ" knots is considered a high speed overrun.
ASMP 3.2.2-1 entails SASP 1.1-6: FAIL; witness: `high_speed_overrun=False, xyz_overrun_threshold=0, groundspeed_overrun_kt=0`
SASP 1.1-6 entails ASMP 3.2.2-1: PASS

Summary: 7 equivalent, 3 not equivalent.